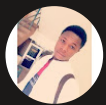


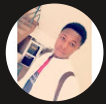


Fetch Data with AWS Lambda



Ahmed Tetteh





Introducing Today's Project!

In this project, I will demonstrate a data logic of my Three tier web project, by creating a creating a serverless function that restrieves user data from a database.

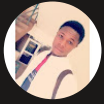
Tools and concepts

Services I used were Lambda and DynamoDB. Key concepts I learnt include Lambda functions how to set up a DynamoDB database. creating, configuring and testing a Lambda function.

Project reflection

This project took me approximately 1 hour 20 mins. The most challenging part was learning how integrate Lambda with other services using an Execution role and the right permissions attached. It was most rewarding to tighten up the security around my Lambda function, but still seein it perform exactly as intended.

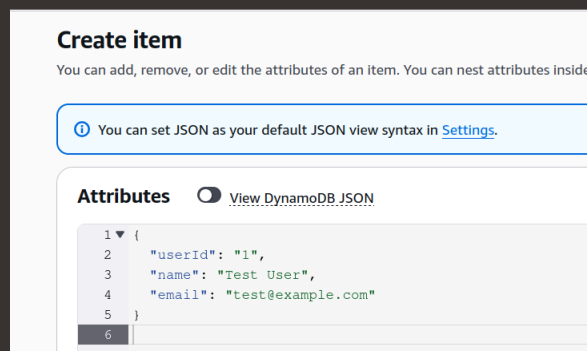
I chose to do this project today because it showed me how serverless architectures work in the real-world.



Project Setup

To set up my project, I created a database table and a simple primary key. The simple primary key consists of the partition key, which is a hash value used to retrieve items from the table. This is how DynamoDB spreads the data across multiple instances to perform efficient querying.

In my DynamoDB table, I added a user data with some few more attributes, such as a name and an email. DynamoDB is schemaless, which means you don't need to know all the attributes before hand to create your data or populate the table. You can add different attributes and different items can belong to different sets of attributes.



Create item

You can add, remove, or edit the attributes of an item. You can nest attributes inside of other attributes.

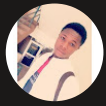
ⓘ You can set JSON as your default JSON view syntax in [Settings](#).

Attributes ☐ View DynamoDB JSON

```
1 {  
2   "userId": "1",  
3   "name": "Test User",  
4   "email": "test@example.com"  
5 }  
6
```

AWS Lambda

AWS Lambda is a service that runs code without the need for developers without provisioning or managing servers . I'm using Lambda in this project to trigger a function(code) that retrieves data from my DynamoDB table.

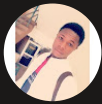


AWS Lambda Function

My Lambda function has an execution role, which is simply the role it needs to perform some basic functions. By default, the role grants Lambda the permission to write logs to CloudWatch.

My Lambda function will interact with DynamoDB. It takes in a UserId, checks the DynamoDB table for a corresponding ID, and returns the data from the table. Else, it returns an error message or data not found.

The code uses AWS SDK, which is a library of pre-built code that allows developers to interact with AWS services. My code uses SDK to interact with DynamoDB.



Lambda > Functions > RetrieveUserData

+ Add trigger + Add destination Last modified 1 minute ago

Code Test Monitor Configuration Aliases Versions

Code source info Open in Visual Studio Code Upload

EXPLORER

- RETRIEVEUSERDATA
 - index.mjs

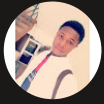
DEPLOY Current

Deploy (Ctrl+Shift+U) Test (Ctrl+Shift+T)

TEST EVENTS [NONE SELECTED]
+ Create new test event

```
1 import { DynamoDBClient } from "@aws-sdk/client-dynamodb";
2 import { DynamoDBDocumentClient, GetCommand } from "@aws-sdk/lib-dynamodb";
3
4 const ddbClient = new DynamoDBClient({ region: 'us-east-1' }); // Make sure to replace this with your actual AWS region
5 const ddb = DynamoDBDocumentClient.from(ddbClient);
6
7 async function handler(event) {
8   const userId = String(event.userId); // Make sure to extract userId from the event
9   const params = {
10     TableName: 'UserData',
11     Key: { userId }
12   };
13
14   try {
15     const command = new GetCommand(params);
16     const data = await ddb.send(command); // Log the raw response from DynamoDB
17     console.log("DynamoDB Response:", JSON.stringify(data));
18     const { Item } = data;
19     if (Item) {
20       console.log("User data retrieved:", Item);
21       return Item;
22     } else {
23       console.log("No user data found for userId:", userId);
24       return null;
25     }
26   } catch (err) {
27     console.error("Unable to retrieve data:", err);
28     return err;
29   }
30 }
```

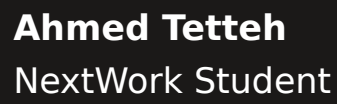
No changes to deploy



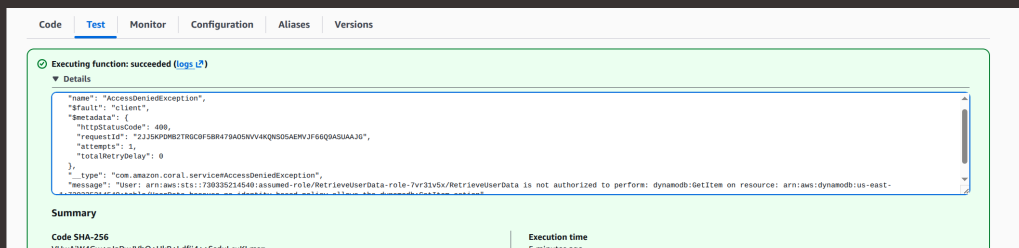
Function Testing

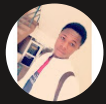
To test whether my Lambda function works, I ran a sample test event with a `UserId`. The test is written in JSON format, making it very easy for our Lambda function to interpret, regardless of the runtime we choose. If the test is successful, I'd see "Executing function: succeeded" message.

The test displayed a 'success' because the function itself run without any errors in the code, but the function's response was actually an error message because my Lambda Function's execution role doesn't contain the permissions to access the DynamoDB table, hence, it was blocked by DynamoDB.



nextwork.org



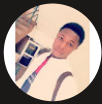


Function Permissions

To resolve the `AccessDenied` error, I reviewed the error log to determine the exact type of permissions my Lambda Function will need to interact with DynamoDB.

There were four DynamoDB permission policies I could choose from, but I didn't pick `"AWSLambdaDynamoDBExecutionRole"`, `"AWSLambdaInvocation-DynamoDB"` and `"AmazonDynamoDBFullAccess"` because they didn't correspond to the right permissions my Lambda Function needed to do its job, i.e, `"dynamodb:GetItem"`.

I also didn't pick `"AmazonDynamoDBFullAccess"` because full access does not comply with the least privilege rule in AWS for a resource or entity, making it less secure if compromised. `AmazonDynamoDBReadOnlyAccess` was the right choice because it contained the exact permissions my Lambda Function needed to its job, without being compromising my DynamoDB table. Also, my Lambda Function just needs to read and retrieve items from the table, not write.



RetrieveUserData-role-7wr31v5x > Add Permissions

Other permissions policies (1/1153)

Q Dyna X Filter by Type All types

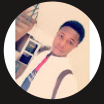
	Policy name	Type
☐	AmazonDynamoDBFullAccess	AWS managed
☐	AmazonDynamoDBFullAccess_v2	AWS managed
☐	AmazonDynamoDBFullAccesswithDataPipeline	AWS managed
☒	AmazonDynamoDBReadOnlyAccess	AWS managed

AmazonDynamoDBReadOnlyAccess

Provides read only access to Amazon DynamoDB via the AWS Management Console.

```
11 {
12   "cloudwatch:DescribeAlarms",
13   "cloudwatch:DescribeAlarmsForMetric",
14   "cloudwatch:GetMetricStatistics",
15   "cloudwatch:ListMetrics",
16   "cloudwatch:GetMetricData",
17   "datapipeline:DescribeObjects",
18   "datapipeline:DescribePipelines",
19   "datapipeline:GetPipelineDefinition",
20   "datapipeline:ListPipelines",
21   "datapipeline:QueryObjects",
22   "dynamodb:BatchGetItem",
23   "dynamodb:Describe*",
24   "dynamodb:List*",
25   "dynamodb:GetItem",
26   "dynamodb:DescribeStatus",
27   "dynamodb:BatchWrite",
28   "dynamodb:Query",
29   "dynamodb:Scan",
30   "dynamodb:PartiQLSelect",
31   "dynamodb:Describe*",
32 }
```

☐	AWSLambdaDynamoDBExecutionRole	AWS managed
☐	AWSLambdaInvocation-DynamoDB	AWS managed

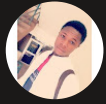


Final Testing and Reflection

To validate my new permission settings, I re-run my test event again. The results were succesful, as expected because my Lambda Function now had the right permissions to retrieve items from the table.

Web apps are a popular use case of using Lambda and DynamoDB. For example, I could use Lambda to retrieve cart items from a DynamoDB table on a product site or even get product information.



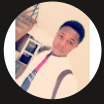


Enhancing Security

For my project extension, I challenged myself to tighten the security of my DynamoDB table by creating an Inline policy for my Lambda Function.

To create the permission policy, I used an Inline policy because it allows me to provides even more granular control over a resource, eliminating all other unnecessary permissions the resource will not use - making it more secure.

I verified that my Lambda function still works by re-running a test event to validate if the Function still works as expected.



[RetrieveUserData-role-7vr31v5x](#) > Create policy

Specify permissions [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "VisualEditor0",  
6       "Effect": "Allow",  
7       "Action": "dynamodb:GetItem",  
8       "Resource": "arn:aws:dynamodb:us-east-1:730335214540:table/UserData"  
9     }  
10  ]  
11 }  
12
```




nextwork.org

The place to learn & showcase your skills

Check out nextwork.org for more projects

